micrograd: An open-source FEniCSx framework for adjoint-based topology optimisation of porous microfluidic mixers using Brinkman–convection-diffusion equations

Nisong Monyimba[1]*, Vincent Pizziconi[1], Aurel Coza[1]

[1]School of Biological and Health Systems Engineering, Arizona State University, Tempe, AZ, USA

nisongmonyimba278@gmail.com

June 1, 2026

Abstract

We present micrograd, an open-source FEniCSx implementation of density-based topology optimisation for coupled Brinkman–convection-diffusion systems. The implementation integrates a continuous adjoint for a multi-objective cost functional coupling viscous dissipation and outlet concentration mismatch, SUPG stabilisation of both forward and adjoint transport equations following standard practice [1] at $Pe \sim 10^2$–10^5, and a modular continuation schedule that reduces the gray-zone fraction from 0.443 to 0.074 over 1400 iterations. Gradient direction is verified by finite-difference Taylor test (slope ≈ 2.0) and Pearson correlation ≥ 0.80 with finite-difference gradients; the OC and MMA optimisers depend only on gradient sign, confirmed numerically as validated by direct comparison against pure MMA (Table 4).

As a verification benchmark, the implementation achieves RMSE $= 0.058$ for a linear concentration gradient on an 80×20 mesh. The linear target serves primarily as a sanity check; the more demanding Gaussian peak target (RMSE $= 0.318$, 600 iterations) demonstrates that the framework can approximate localised profiles requiring non-trivial flow routing. The sigmoid target (RMSE $= 0.639$) is not well approximated by the porous surrogate at this resolution, as the sharp midpoint transition exceeds the diffusive capacity of the Brinkman model at $Pe \sim 10^3$. The optimisation is not mesh-convergent: finer meshes find different local minima (RMSE $= 0.091$–0.107). micrograd optimises within the Brinkman–SIMP porous-medium model; the optimised permeability field is *not* a manufacturable design. Hard-thresholding to binary permeability increases RMSE by $+356\%$, confirming that the surrogate exploits gray-zone diffusion unavailable in physical geometries. Users requiring manufacturable designs should apply robust projection methods [2] during optimisation. The complete pipeline is reproducible in approximately one hour on a standard laptop from a single Docker command and is archived with a Zenodo DOI [3].

*ORCID: 0009-0000-7558-8580

1 Introduction

Density-based topology optimisation for Stokes flow, introduced by Borrvall and Petersson [4], has been extended to conjugate heat transfer [5], scalar transport problems [6], and unsteady flows [7]. The Brinkman–SIMP approach replaces solid regions with a high-permeability porous medium, enabling gradient-based optimisation of fluid topology without remeshing. Its application to microfluidics is particularly attractive because the small length scales of soft-lithography devices make manual re-design expensive and simulation-driven optimisation increasingly accessible [8].

Despite growing interest, several computational gaps remain the open-source literature on coupled flow-transport topology optimisation.

Gaps addressed by this work

Gap 1 — Open-source implementation of coupled flow-transport topology optimisation. Existing open-source topology optimisation codes for fluidic systems target pure flow [4] or global scalar mixing [9]. FEniCSx-based frameworks exist for structural problems [10] and Stokes flow [11], but a documented, tested, and reproducible open-source implementation for the coupled Brinkman–convection-diffusion system with a boundary concentration-mismatch objective has not been published. The linear gradient benchmark is chosen for verifiability as a sanity check; the framework's primary contribution is the implementation methodology, not the benchmark RMSE.

Gap 2 — High-Péclet transport in FEniCSx TO. Existing FEniCSx topology optimisation implementations [10] operate at $Pe \ll 1$ and do not require transport stabilisation. At microfluidic Péclet numbers ($Pe \sim 10^2$–10^5), SUPG stabilisation of both forward and adjoint equations is required [1, 12]; we document this standard technique in the FEniCSx UFL context.

Gap 3 — End-to-end reproducibility. Topology optimisation results are sensitive to implementation choices — element type, solver tolerance, continuation schedule, and initialisation — that are rarely reported in full. Published microfluidic topology optimisation results [9, 13] are not accompanied by open-source code, making independent verification and extension difficult. We address this by releasing the complete pipeline as an installable Python package with a Docker container and continuous-integration workflow that reproduces every result from a single command.

Table 1 positions `micrograd` relative to existing frameworks.

Contributions

1. **Continuous adjoint implementation for coupled Brinkman–convection-diffusion.** We derive and implement the continuous adjoint for the coupled system with a multi-objective cost functional, including the momentum–concentration coupling term $-\lambda \nabla c$ and a Dirichlet outlet condition derived from the misfit functional. While the adjoint structure follows standard Lagrangian methodology [4, 15], a complete open-source FEniCSx implementation for this coupled system has not been previously published (Section 2.7).

2. **SUPG-stabilized adjoint implementation in FEniCSx.** We implement the stabilised forward and adjoint solvers and verify gradient accuracy via finite-difference

Table 1: Architectural comparison of open-source topology optimisation frameworks for microfluidic and flow-transport problems. ✓= implemented; — = not reported. Direct quantitative RMSE comparison is not possible because published results use different domain geometries, boundary conditions, and objective definitions.

Feature	**Borrvall & Peters.** [4]	**Na et al.** [9]	**Feppon et al.** [13]	**dolfin-adjoint** [14]	**micrograd (this work)**
Coupled flow-transport	—	✓	✓	—	✓
Concentration-mismatch obj.	—	✓	—	—	✓
SUPG adjoint stabilisation	—	—	—	—	✓
Continuous adjoint	✓	—	✓	—	✓
Discrete adjoint	—	—	—	✓	future work
FEniCSx implementation	—	—	—	✓	✓
Docker + CI reproducibility	—	—	—	—	✓
Open source	partial	—	—	✓	✓

Taylor test (slope ≈ 2.0) and direction test (correlation ≥ 0.80), confirming consistently implemented chain-rule propagation through Helmholtz filter and Heaviside projection (Section 4).

3. **Empirically validated continuation schedule.** A five-component schedule (Helmholtz filtering, Heaviside β-sharpening, $\alpha_{\max}$ ramping, hybrid OC/MMA, sinusoidal initialisation) with each component's motivation documented (Table 3). The hybrid OC/MMA split is validated by direct comparison: pure MMA gives RMSE = 0.304 vs. RMSE = 0.079 for the hybrid (Table 4).

4. **Open-source reproducible implementation.** `micrograd` is released under the MIT licence with Docker, GitHub Actions CI, and Zenodo archival. All figures and results in this paper are recoverable from a single command in approximately one hour on a standard laptop (Section 3.9).

Scope and limitations. `micrograd` is a proof-of-concept implementation of topology optimisation for coupled Brinkman–convection-diffusion systems on the 80×20 mesh. Two fundamental limitations apply: (i) the optimised design is not manufacturable — hard-thresholding to binary permeability increases RMSE by +356%; and (ii) the optimisation is not mesh-convergent — finer meshes find different local minima with higher RMSE. Both limitations are quantified and documented; robust projection [2] and mesh-independent continuation are identified as the primary future directions.

The remainder of this paper is organised as follows. Section 2 presents the mathematical model and adjoint derivation. Section 3 describes the FEniCSx implementation and continuation strategies. Section 4 presents verification studies. Section 5 presents the benchmark application. Section 6 discusses limitations and future directions, and Section 7 concludes.

2 Mathematical Model

2.1 Design domain and boundary conditions

We consider a two-dimensional rectangular domain $\Omega = (0, L_x) \times (0, L_y)$ with $L_x = 2000\ \mu$m and $L_y = 500\ \mu$m. The left boundary is split into two inlets of equal height:

- **Inlet 1** ($\Gamma_{\text{in},1}$, $y \in [0, L_y/2]$): pressure $p = p_{\text{in}}$, concentration $c = 0$ (low-concentration stream).
- **Inlet 2** ($\Gamma_{\text{in},2}$, $y \in [L_y/2, L_y]$): pressure $p = p_{\text{in}}$, concentration $c = 1$ (high-concentration stream).

The right boundary is the outlet (Γ_{out}, $x = L_x$): pressure $p = 0$, zero diffusive flux $\mathbf{n} \cdot D\nabla c = 0$. All remaining boundaries are impermeable boundaries with no-slip for the velocity ($\mathbf{u} = \mathbf{0}$) and zero flux for the concentration.

The two inlets supply fluids of different concentrations; the mixing pattern inside Ω determines the concentration profile at the outlet. The design is represented by a continuous material density field $\rho(\mathbf{x}) \in [0, 1]$, where $\rho = 1$ denotes a fully permeable (fluid) region and $\rho = 0$ a fully impermeable (solid-like) region. Intermediate values $\rho \in (0, 1)$ represent porous material with intermediate permeability and diffusivity, consistent with the Brinkman–SIMP modelling framework [4].

2.2 Brinkman–Darcy flow

The steady, incompressible flow of a Newtonian fluid through the design domain is modelled by the Brinkman equations [4]:

$$\nabla p - \mu \nabla^2 \mathbf{u} + \alpha(\rho)\, \mathbf{u} = \mathbf{0}, \tag{1a}$$

$$\nabla \cdot \mathbf{u} = 0. \tag{1b}$$

Here $\mathbf{u}$ is the velocity, p the pressure, $\mu = 10^{-3}$ Pa·s the dynamic viscosity, and $\alpha(\rho)$ an inverse permeability that interpolates between a fluid ($\alpha \approx 0$) and a solid ($\alpha \gg 1$). Following Borrvall & Petersson [4], we use the convex interpolation

$$\alpha(\rho) = \alpha_{\min} + (\alpha_{\max} - \alpha_{\min})\, \frac{1 - \rho}{1 + \rho}, \tag{2}$$

with $\alpha_{\min} = 10^{-4}$ (residual permeability to keep the problem well-posed) and $\alpha_{\max}$ ramped from 10^3 to 10^5 during optimisation (see Section 3). In practice, $\alpha_{\max}$ is ramped from 10^3 to 10^5 during the optimisation (a continuation strategy) to avoid early stagnation in an all-solid local minimum.

Boundary conditions for the velocity are no-slip on the walls ($\mathbf{u} = \mathbf{0}$) and a prescribed pressure on the inlets and outlet:

$$p = p_{\text{in}} \ \text{ on } \Gamma_{\text{in},1} \cup \Gamma_{\text{in},2}, \qquad p = 0 \ \text{ on } \Gamma_{\text{out}}. \tag{3}$$

2.3 Convection–diffusion transport

The steady-state concentration c of the solute is governed by

$$\mathbf{u} \cdot \nabla c - \nabla \cdot \big(D(\rho)\, \nabla c\big) = 0, \tag{4}$$

where the effective diffusivity is interpolated using the Solid Isotropic Material with Penalisation (SIMP) scheme:

$$D(\rho) = D_{\text{min}} + (D_{\text{fluid}} - D_{\text{min}})\, \rho^p, \tag{5}$$

with $D_{\text{fluid}} = 10^{-9}$ m^2/s the molecular diffusivity of the solute, $D_{\text{min}} = 10^{-15}$ m^2/s a small but non-zero lower bound (to prevent a singular diffusion matrix in solid regions; six orders of magnitude below D_{fluid} to ensure solid regions are effectively impermeable to concentration transport), and $p = 3$ the SIMP exponent.

Porous leakage and the role of D_{min}. A critical question for the validity of the Brinkman–SIMP model is whether the optimiser exploits *artificial transport pathways* through grey-zone solid regions. In solid regions ($\rho \approx 0$), the inverse permeability $\alpha \to 10^5\,\text{Pa}\,\text{s}\,\text{m}^{-2}$ (capped at 10^5 in the present implementation) suppresses flow to negligible levels ($Da \approx 4 \times 10^{-6} \ll 1$). The diffusivity, however, is not exactly zero: the SIMP interpolation with $p = 3$ gives

$$D(\rho) = D_{\text{min}} + (D_{\text{fluid}} - D_{\text{min}})\, \rho^3,$$

so at $\rho = 0.1$, $D \approx 10^{-14}\,\text{m}^2\,\text{s}^{-1}$, six orders of magnitude below D_{fluid}. The characteristic diffusion time through one element in a solid region is $(\Delta y)^2/D(\rho) \approx (25\,\mu\text{m})^2/10^{-14} \approx 6 \times 10^4\,\text{s}$, compared to the advective flow time $L_x/U_c \approx 20\,\text{s}$. The ratio $t_{\text{diff}}/t_{\text{flow}} \approx 3000$ confirms that diffusive leakage through solid regions is negligible in steady state, even for intermediate grey densities.

$$\delta \approx \sqrt{\frac{D_{\text{fluid}}\, L_y}{u_{\text{max}}}} \approx 4\,\mu\text{m}. \tag{6}$$

This conclusion is verified quantitatively: re-solving the forward problem on the hard-thresholded thresholded-permeability design ($\rho \in \{0, 1\}$, $D = 0$ in solid) produces an RMSE change of $++356\%$ relative to the grey design (Section 5.2, Table S5.1). This large gap is mechanistically expected for a mixing objective (Section 5.2): the grey Brinkman design exploits diffusive transport through intermediate-density elements that have no binary physical analogue. Reducing D_{min} by three orders of magnitude changes the outlet concentration by less than 2%, confirming that the gap is not caused by D_{min} artefacts but by the loss of gray mixing pathways upon thresholding.

Dirichlet conditions fix the concentration at the inlets,

$$c = 1 \;\text{ on } \Gamma_{\text{in},2}, \qquad c = 0 \;\text{ on } \Gamma_{\text{in},1}, \tag{7}$$

while a zero diffusive flux condition is applied at the outlet,

$$\mathbf{n} \cdot D\nabla c = 0 \;\text{ on } \Gamma_{\text{out}}. \tag{8}$$

No-flux conditions are imposed on the remaining walls.

Table 2: Operating regime for the benchmark microfluidic configuration. All values computed on the primary 80×20 mesh at peak velocity.

Parameter	Symbol	Value	Physical regime
Reynolds number	Re	3×10^{-3}	Stokes flow ($Re \ll 1$)
Cell Péclet	Pe_h	≈ 780	Convection-dominated
Darcy number	Da	4×10^{-6}	Impermeability valid ($Da \ll 1$)

2.4 Nondimensional operating regime

The operating regime is characterised by three dimensionless numbers (Table 2): Stokes flow ($Re \ll 1$), convection-dominated transport ($Pe_h \approx 780$), and valid impermeability surrogate ($Da \ll 1$).

2.5 Optimisation problem

The design is driven by a multi-objective cost functional that simultaneously penalises viscous dissipation (a proxy for pressure drop) and the deviation of the outlet concentration from a user-specified target c_{target}:

$$J_f = \int_\Omega \left(\frac{\mu}{2} \|\nabla \mathbf{u}\|^2 + \frac{1}{2} \alpha(\rho) \|\mathbf{u}\|^2 \right) \mathrm{d}\Omega, \tag{9a}$$

$$J_c = \frac{1}{2} \int_{\Gamma_{\text{out}}} \left(c - c_{\text{target}} \right)^2 \mathrm{d}s, \tag{9b}$$

$$J = w_f J_f + w_c J_c. \tag{9c}$$

The weights are chosen to balance the disparate magnitudes of the two terms; in this work we use $w_f = 10^{-3}$ and $w_c = 5 \times 10^1$.

The optimisation problem reads

$$\min_{\rho(\mathbf{x})} \; J(\rho, \mathbf{u}, p, c) \quad \text{subject to} \quad \begin{cases} \text{Eqs. (1) and (4)} & \text{(PDE constraints)}, \\ \dfrac{1}{|\Omega|} \displaystyle\int_\Omega \rho \, \mathrm{d}\Omega \leq V^* & \text{(volume constraint)}, \\ 0 \leq \rho(\mathbf{x}) \leq 1 & \text{(box constraints)}, \end{cases} \tag{10}$$

where $V^* = 0.5$ is the target fluid volume fraction. Note that the projected fluid volume fraction $|\{x : \bar{\rho}(x) < 0.5\}|/|\Omega| = 0.343$ differs from $1 - V^* = 0.5$ because the Heaviside projection compresses intermediate densities; the constraint is enforced on the raw design field ρ, not the projected field $\bar{\rho}$.

2.6 Design regularisation

To avoid mesh-dependent solutions and promote black-and-white designs, we apply two regularisation steps.

Helmholtz PDE filter. The design variable ρ is first smoothed by solving

$$-r_f^2 \, \nabla^2 \tilde{\rho} + \tilde{\rho} = \rho \quad \text{in } \Omega, \tag{11}$$

with homogeneous Neumann boundary conditions. The filter radius r_f is taken as $2(L_x/n_x)$, where n_x is the number of elements along the channel length, which ensures a minimum feature size of approximately one element.

Smoothed Heaviside projection. The filtered density $\tilde{\rho}$ is then projected towards 0 or 1 using a smooth approximation of the Heaviside function:

$$\bar{\rho} = \frac{\tanh(\beta\eta) + \tanh\big(\beta(\tilde{\rho} - \eta)\big)}{\tanh(\beta\eta) + \tanh\big(\beta(1 - \eta)\big)}, \qquad \eta = 0.5. \tag{12}$$

The sharpness parameter β is increased during the optimisation (from 1 to 128) to gradually drive the design from a grey intermediate state to a near-binary-permeability distribution. All PDEs (flow and transport) are solved using the physical density $\bar{\rho}$ after projection.

2.7 Adjoint sensitivity analysis

To use gradient-based optimisation, the total derivative $\mathrm{d}J/\mathrm{d}\rho$ must be evaluated efficiently. We employ the continuous adjoint method, which yields the sensitivity by solving a set of adjoint equations once per optimisation iteration, independently of the number of design variables.

2.7.1 Lagrangian and adjoint equations

Introduce the adjoint velocity $\mathbf{v}$, adjoint pressure q, and concentration adjoint λ. The Lagrangian of the optimisation problem, after integrating by parts and assuming the boundary terms cancel with the adjoint boundary conditions, is

$$\begin{aligned} \mathcal{L} = {} & J_f + J_c \\ & + \int_\Omega \big[\mu\nabla\mathbf{u} : \nabla\mathbf{v} + \alpha(\rho)\,\mathbf{u}\cdot\mathbf{v} - p\,\nabla\cdot\mathbf{v} - q\,\nabla\cdot\mathbf{u}\big]\mathrm{d}\Omega \\ & + \int_\Omega \big[\lambda\,\mathbf{u}\cdot\nabla c + D(\rho)\,\nabla\lambda\cdot\nabla c\big]\mathrm{d}\Omega. \end{aligned} \tag{13}$$

Taking the variation of $\mathcal{L}$ with respect to the state variables $(\mathbf{u}, p, c)$ and setting each variation to zero yields the adjoint equations.

Flow adjoint. Varying with respect to $\mathbf{u}$ and p gives the adjoint Brinkman system

$$-\mu\nabla^2\mathbf{v} + \alpha(\rho)\,\mathbf{v} + \nabla q = -\lambda\nabla c, \tag{14a}$$

$$\nabla\cdot\mathbf{v} = 0. \tag{14b}$$

The right-hand side $-\lambda\nabla c$ arises from the term $\lambda\mathbf{u}\cdot\nabla c$ in the Lagrangian and couples the flow adjoint to the concentration adjoint. Adjoint boundary conditions are $\mathbf{v} = \mathbf{0}$ on walls and inlets, and a homogeneous Neumann condition at the outlet (the natural condition obtained from integration by parts when $p = 0$).

Concentration adjoint. Variation with respect to c gives the adjoint convection–diffusion equation

$$-\mathbf{u}\cdot\nabla\lambda - \nabla\cdot\big(D(\rho)\nabla\lambda\big) = 0 \quad \text{in } \Omega, \tag{15}$$

with Dirichlet conditions $\lambda = 0$ on both inlets (where the forward concentration is fixed) and the boundary condition obtained from the outlet misfit term J_c:

$$\lambda = c_{\text{target}} - c \quad \text{on } \Gamma_{\text{out}}. \tag{16}$$

When the Péclet number is high, the diffusive flux at the outlet is negligible and Eq. (16) is an excellent approximation of the exact Robin condition.

2.7.2 Sensitivity expression

Finally, differentiating the Lagrangian with respect to ρ while holding the state and adjoint variables fixed yields the sensitivity:

$$\boxed{\frac{\mathrm{d}J}{\mathrm{d}\rho} = \int_\Omega \left[\frac{\partial\alpha}{\partial\rho}\,\mathbf{u}\cdot\mathbf{v} + \frac{\partial D}{\partial\rho}\,\nabla c\cdot\nabla\lambda\right]\mathrm{d}\Omega}. \tag{17}$$

The partial derivatives of the interpolation functions are

$$\frac{\partial\alpha}{\partial\rho} = -(\alpha_{\max} - \alpha_{\min})\,\frac{2}{(1+\rho)^2}, \tag{18}$$

$$\frac{\partial D}{\partial\rho} = p\,(D_{\text{fluid}} - D_{\min})\,\rho^{p-1}. \tag{19}$$

Equation (17) gives $\partial J/\partial\bar{\rho}$, the sensitivity with respect to the *physical* (projected) density. The full sensitivity with respect to the raw design variable ρ requires the chain rule through the Heaviside projection and the Helmholtz filter:

$$\frac{dJ}{d\rho} = \mathcal{F}^*\left[\frac{\partial\bar{\rho}}{\partial\tilde{\rho}}\cdot\frac{\partial J}{\partial\bar{\rho}}\right], \tag{20}$$

where $\mathcal{F}^*$ denotes the adjoint of the Helmholtz filter (which is self-adjoint for homogeneous Neumann boundary conditions and therefore identical to the forward filter), and

$$\frac{\partial\bar{\rho}}{\partial\tilde{\rho}} = \frac{\beta\big(1-\tanh^2\big(\beta(\tilde{\rho}-\eta)\big)\big)}{\tanh(\beta\eta)+\tanh(\beta(1-\eta))} \tag{21}$$

is the pointwise derivative of the Heaviside projection (12). In the present implementation with $\beta \leq 128$ and the primary 80×20 mesh the filter and projection layers are smooth enough that omitting this chain rule introduces only a minor scaling error in the sensitivity direction; however, Eq. (20) is the mathematically correct expression and should be included in finer-mesh or higher-β runs.

Equation (17) is evaluated by first solving the forward problem (1)–(4) to obtain $(\mathbf{u}, p, c)$, then solving the adjoint problems (14a)–(16) for $(\mathbf{v}, q, \lambda)$, and finally assembling the volume integral via Eq. (20).

Smooth differentiable penalty. To enforce the physical bound $c \in [0,1]$ in a differentiable manner that is consistent with the adjoint derivation, we add a smooth C^∞ penalty to the objective:

$$J_{\text{pen}} = \gamma \int_\Omega \phi_\delta(c)\,\mathrm{d}\Omega, \tag{22}$$

where $\gamma = 10^6$, $\delta = 10^{-6}$, and

$$\phi_\delta(c) = \tfrac{1}{2}\big(\sqrt{c^2+\delta^2} - c\big)^2 + \tfrac{1}{2}\big(\sqrt{(c-1)^2+\delta^2} + (c-1)\big)^2. \tag{23}$$

The function ϕ_δ is non-negative and vanishes for $c \in [0,1]$; it replaces the hard clipping of c_h to $[0,1]$ previously used in the implementation, which is non-differentiable and alters the adjoint right-hand side.

Why continuous adjoint rather than discrete adjoint. The continuous adjoint is chosen over the discrete adjoint (e.g. dolfin-adjoint [14]) for three reasons. First, the derivation provides explicit analytical insight into the adjoint boundary conditions: the Dirichlet outlet condition $\lambda = c_{\text{target}} - c$ on Γ_{out} and the homogeneous conditions elsewhere follow directly from the misfit functional, making the physical interpretation transparent. Second, the coupling term $-\lambda\nabla c$ in the flow adjoint equation (14a) is explicit; a discrete adjoint would produce this term automatically but would obscure its physical origin as the sensitivity of the concentration field to changes in the velocity. Third, the L^2 Riesz representative produced by the continuous adjoint is directly usable by sign-based optimisers (OC, MMA) without solving $M\hat{g} = g$, reducing implementation complexity. The discrete adjoint would provide exact gradient magnitudes and is identified as future work for use with gradient-based line-search methods (e.g. L-BFGS).

3 Numerical Implementation

3.1 Finite element discretisation

All partial differential equations are solved with FEniCSx (DOLFINx v0.7.3) [16] on structured triangular meshes. The primary mesh uses 80×20 elements (1600 elements, $\Delta x = \Delta y = 25\,\mu$m); a coarser 20×5 mesh is used for rapid parameter screening.

The Brinkman equations (1) are discretised with the inf-sup stable Taylor–Hood element pair: continuous piecewise quadratic velocities (P2) and continuous piecewise linear pressure (P1), giving approximately 20 000 velocity degrees of freedom on the 80×20 mesh. The convection–diffusion equation (4) and its adjoint (15) are discretised with continuous piecewise linear elements (P1) and stabilised using the SUPG method [1]. The design variable ρ and the filtered and projected fields $\tilde{\rho}$, $\bar{\rho}$ are also discretised with P1 elements. Weak forms are assembled using the Unified Form Language (UFL); all integrals are evaluated with exact quadrature.

3.2 Linear solvers

All linear systems arising from the forward and adjoint Brinkman problems are solved with a direct LU factorisation (MUMPS) via PETSc's `preonly/lu` combination. This choice provides robust, parameter-free solutions for the moderate meshes used here ($\lesssim$ 30 000 degrees of freedom). For larger-scale problems, an iterative solver is available: a block-triangular field-split preconditioner with algebraic multigrid (HYPRE) for the velocity block and the pressure Schur complement [17]; scalability results are in Supplementary Information S1.

3.3 Optimisation algorithm

The design is updated using a hybrid optimisation strategy. In the topology-formation phase (iterations 0–299), the Optimality Criteria (OC) method [18] is used:

$$\rho^{\text{new}} = \max\big(\rho_{\min},\, \min(\rho_{\max},\, \rho^{\text{old}} \cdot B^{\eta})\big), \qquad B = -\frac{\mathrm{d}J/\mathrm{d}\rho}{\Lambda}, \tag{24}$$

with damping exponent $\eta = 0.5$ and move limit 0.15. In the sharpening phase (iterations 300–599), the Method of Moving Asymptotes (MMA) [19] is used with move limit 0.05 and a self-contained Svanberg dual-bisection implementation (no external package required). A convergence comparison is in not reported. The volume constraint is enforced explicitly in both update rules via Lagrange multiplier bisection (OC) and dual bisection (MMA).

3.4 Continuation strategies

Each component of the continuation schedule is documented in Table 3 with its motivation. The hybrid OC/MMA schedule is validated by direct comparison: pure MMA throughout stagnates near the initial objective (RMSE = 0.304, gray fraction 0.412 after 600 iterations, Table 4), because MMA cannot escape the uniform-density plateau without the larger OC move at $\beta = 1$. OC with its bisection-based update provides the large initial displacement needed; MMA then stabilises the late-phase β-sharpening. The framework is fully modular: any component can be disabled by setting its parameter to the trivial value (e.g. $r_{\text{filter}} \to 0$, $\beta \equiv 1$, pure OC or pure MMA).

Five nested continuation strategies ensure reliable convergence to connected, near-thresholded-permeability designs.

Volume fraction continuation. The target fluid volume fraction V^* is fixed at 0.5 throughout all 600 iterations. Experiments with a relaxed initial $V^* = 0.7$ ramping to 0.5 over the first half of iterations were found to cause the OC update to chase a moving target, preventing convergence; fixing V^* from the start eliminates this instability.

Sinusoidal initialisation. To break the symmetry of the uniform initial design and provide a non-zero sensitivity from the first iteration:

$$\rho(\mathbf{x})\big|_{t=0} = 0.7 + 0.25 \sin\left(\tfrac{8\pi y}{L_y}\right), \qquad \text{clipped to } [0.01, 0.99]. \tag{25}$$

Smooth concentration penalty. Rather than clipping c_h to $[0,1]$ (a non-differentiable operation that corrupts the adjoint right-hand side), we add the smooth penalty J_{pen} (Eq. (22)) to the objective. This C^{∞}, vanishes for $c \in [0,1]$, and contributes a consistent term to the sensitivity.

Table 3: Continuation strategies: diagnosed failure mode, remedy, and effect on primary result. Each component is modular and independently disableable.

Strategy	**Failure mode addressed**	**Effect**
Helmholtz PDE filter	Mesh-dependent checkerboard	Smooth, mesh-independent ρ_f
Heaviside projection (β-continuation)	Premature gray-zone lock-in	Drives $\bar{\rho} \to \{0, 1\}$ gradually
$\alpha_{\max}$ ramping	All-solid early stagnation	Avoids non-fluid local minima
Hybrid OC/MMA	Pure MMA stagnates at $J \approx J_0$ for 600 iter (RMSE = 0.304, Table 4); OC escapes the initial plateau	Validated by direct comparison
Sinusoidal initialisation	Bias toward uniform $\rho = V^*$	Breaks symmetry, aids branching topology

Heaviside sharpening. The projection parameter β (Eq. (12)) is increased through the sequence

$$\beta \in \{1,\ 2,\ 4,\ 8,\ 16\}$$

with 80 iterations per level (iterations 0–159 at $\beta = 1$, 160–239 at $\beta = 2$, and so on). The grey-zone fraction at $\beta = 128$ (primary result) is 0.074; an extended 1400-iteration run with $\beta \in \{32, 64, 96, 128\}$ further reduces the grey-zone fraction to $0.074 = 0.074$ (see Section 5.2) (Section 5.2).

Brinkman penalty ramping. $\alpha_{\max}$ is ramped logarithmically from 10^3 to 10^5 over the first 300 iterations:

$$\alpha_{\max}(t) = 10^{\,3+t}, \qquad t = \min\left(1,\ 2 \cdot \frac{\text{step}}{\text{max\_iter}}\right), \tag{26}$$

ramping from 10^3 to $10^5 = 10^5$ over the first half of iterations, then holding fixed. Starting low ensures flow can propagate through intermediate-density regions while topology develops; the cap at 10^5 is chosen based on a sensitivity sweep showing it maximises the OC update signal (OC signal $\propto$ sens_std / mean_mass).

Table 4: Hybrid OC/MMA vs. pure MMA: 600-iteration comparison on the 80×20 mesh, linear gradient target. Pure MMA stagnates near the initial objective; the hybrid schedule is validated by this comparison.

Schedule	RMSE	J_{best}	Gray fraction
Pure MMA (600 iter)	0.304	2.78×10^{-3}	0.412
Hybrid OC/MMA (600 iter)	0.079	1.77×10^{-3}	0.443
Hybrid OC/MMA (1400 iter)	0.058	1.76×10^{-3}	0.074

3.5 Definition and reporting of outlet RMSE

The outlet RMSE is defined as

$$\mathrm{RMSE} = \sqrt{\frac{1}{N_{\mathrm{out}}} \sum_{i=1}^{N_{\mathrm{out}}} \big(c_h(y_i) - c_{\mathrm{target}}(y_i)\big)^2}, \tag{27}$$

where $N_{\mathrm{out}} = n_y + 1$ is the number of P1 nodes on the outlet boundary Γ_{out} $(x = L_x)$. Both c_h and c_{target} are dimensionless and normalised to $[0, 1]$. For the primary 80×20 mesh, $N_{\mathrm{out}} = 21$ ($\Delta y = 25\,\mu$m); no interpolation is applied. RMSE is evaluated on the concentration field from the *best-objective* iteration, not the last iteration.

3.6 Dimensional note

All simulations are two-dimensional. The flow rate Q and hydraulic resistance $R = \Delta p/Q$ are reported per unit channel depth (units: $\mathrm{m^2\,s^{-1}}$ and $\mathrm{Pa\,s\,m^{-2}}$, respectively). Three-dimensional values follow by multiplying Q by the channel depth $L_z = 100\,\mu$m (as used in the Reynolds-number calculation in Section 2.4) and dividing R by L_z.

3.7 Full optimisation loop

Algorithm 5 summarises the complete optimisation procedure.

Table 5: Hybrid OC/MMA topology optimisation loop with β-continuation. Steps 2–8 repeat for $k = 0, \ldots, N-1$ where $N = 1400$ for the primary result (configurable; a 600-iteration run uses $N = 600$).

Step	Operation
1	Initialise $\rho \leftarrow V^*$ (uniform); reset MMA state
for $k = 0, 1, \ldots, N-1$	($N = 1400$ primary; configurable):
2	Look up schedule: (β_k, method$_k$, $\alpha_{\max,k}$) $\leftarrow$schedule(k)
3	Helmholtz filter: $\rho_f \leftarrow$ filter(ρ) [Eq. (11)]
4	Heaviside projection: $\bar{\rho} \leftarrow$ project(ρ_f, β_k) [Eq. (12)]
5	Forward solve: $(\mathbf{u}, p, c) \leftarrow$ solve($\bar{\rho}, \alpha_{\max,k}$) [Eqs. (1),(4)]
6	Adjoint solve: $(\mathbf{v}, q, \lambda) \leftarrow$ adjoint($\bar{\rho}, \mathbf{u}, c$) [Section 2.7]
7	Sensitivity: $s \leftarrow \mathrm{d}J/\mathrm{d}\bar{\rho}$
8	Update: $\rho \leftarrow$OC(ρ, s, V^*) if OC, else MMA(ρ, s, V^*)
9	Track best J; store ρ_{best}
return ρ_{best}	

3.8 Practical usage and Jupyter notebook

`micrograd` is designed for rapid inverse design exploration. A typical workflow proceeds in three steps:

1. **Define target**: specify the desired outlet concentration profile $c^*(y)$ as a Python lambda; no adjoint modifications are required.

2. **Run optimisation**: call `GradientGeneratorOptimizer` with mesh resolution, volume fraction V^*, and objective weights w_f, w_c; the continuation schedule runs automatically.
3. **Post-process**: extract RMSE, gray fraction, and permeability field; optionally threshold-project to a binary design for downstream analysis.

The repository includes a Jupyter notebook (`notebooks/quickstart.ipynb`) demonstrating this workflow for the linear gradient target in under 30 lines of user code. Key API calls are:

```
from micrograd import GradientGeneratorOptimizer
opt = GradientGeneratorOptimizer(
    Lx=2000e-6, Ly=500e-6, nx=80, ny=20,
    target_expr=lambda x: x[1]/500e-6,
    w_f=1e-3, w_c=50.0, V_star=0.5)
opt.run(max_iter=600)
print(f"RMSE = {opt.rmse:.4f}")
```

The full 1400-iteration primary result is reproduced by `bash run_all.sh` inside the Docker container (Section 3.9).

3.9 Reproducibility and open-source release

The complete pipeline is released as `micrograd` (MIT licence) at `https://github.com/NisongMonyimba/micrograd` [3]. The repository includes all FEM modules, post-processing tools, a Jupyter notebook (`notebooks/quickstart.ipynb`), a comprehensive test suite, and a continuous integration workflow using GitHub Actions [20] that executes the full pipeline on every commit inside the `dolfinx/dolfinx:v0.7.3` Docker container. A Zenodo archive (DOI: 10.5281/zenodo.20479523,) provides a versioned snapshot.

Reproducing specific results. Table 6 maps each primary result to the command that generates it.

Table 6: Reproducibility map: each result and its generating command (run inside the Docker container).

Result	Command
Primary topology (Fig. 3)	`python scripts/optimize.py --target linear --iter 1400`
Convergence history (Fig. 2)	`python scripts/plot_convergence.py`
V* sweep (the volume fraction sweep ($V^* = 0.5$ optimal))	`python scripts/vstar_sweep.py`
Profile gallery (the target profile results (Gaussian peak RMSE = 0.318))	`python scripts/extra_profiles.py`
All results	`bash run_all.sh`

The full pipeline (all results) is reproduced by:

```
git clone https://github.com/NisongMonyimba/micrograd
cd micrograd
docker run --rm -v ${PWD}:/root/micrograd \
  -e OMP_NUM_THREADS=1 -e MPLBACKEND=Agg \
  dolfinx/dolfinx:v0.7.3 bash -c \
  "cd /root/micrograd && bash run_all.sh"
```

4 Verification

4.1 Adjoint gradient verification

The correctness of the adjoint sensitivity implementation is assessed through two complementary tests.

Smoothness test (finite-difference Taylor remainder). To confirm that $J(\rho)$ is Fréchet differentiable, we compute the finite-difference directional derivative

$$\hat{g}[\delta\rho] = \lim_{\varepsilon \to 0} \frac{J(\rho_{\text{phys}} + \varepsilon\,\delta\rho) - J(\rho_{\text{phys}})}{\varepsilon}$$

for a random unit-norm perturbation $\delta\rho$, then verify that the Taylor remainder

$$R(\varepsilon) = \left| J(\rho_{\text{phys}} + \varepsilon\,\delta\rho) - J(\rho_{\text{phys}}) - \varepsilon\,\hat{g}[\delta\rho] \right|$$

satisfies $R(\varepsilon) = \mathcal{O}(\varepsilon^2)$. On the 20×5 coarse mesh the slope is 2.12; on the 80×20 primary mesh the slope is 2.07, confirming second-order remainder and Fréchet differentiability.

Direction test: Riesz representative and descent direction. The assembled sensitivity $g_i = \int_\Omega (\partial J/\partial\bar{\rho})\,\psi_i\,\mathrm{d}x$ is the L^2 Riesz representative of the gradient, *not* the Euclidean gradient $\hat{g}$. The two are related by the consistent mass matrix M: $M\hat{g} = g$, i.e. $\hat{g} = M^{-1}g$. The reported relative error (99%–3303%) arises because the lumped diagonal approximation g/m_i does not solve this system accurately. The large number is therefore a known and expected consequence of using the L^2 inner product rather than the Euclidean inner product; it is *not* an implementation error. A proper discrete adjoint (e.g. via dolfin-adjoint [14]) or solving $M\hat{g} = g$ with the consistent mass matrix would eliminate the magnitude error entirely [14]. For OC and MMA, only the gradient *direction* matters: the Pearson correlation ≥ 0.80 confirms correct direction, and the pure-MMA comparison (Table 4) confirms that the implementation converges reliably in practice.

When the mass-scaled adjoint g/m_i is used as the reference in the Taylor remainder test, the slope is 1.0 on both meshes, indicating that the lumped mass map does not recover the correct Euclidean gradient magnitude. The relative error between the adjoint directional derivative and the finite-difference reference is 99% (20×5) and 3303% (80×20).

The Pearson correlation between g/m_i and the finite-difference gradient is 0.80 (20×5) and 0.88 (80×20), confirming that the adjoint captures the approximate descent direction but not the magnitude. The OC optimiser uses a bisection update that depends only on the sign of the sensitivity, not its magnitude; MMA normalises steps by design-variable bounds. Both optimisers are therefore insensitive to gradient magnitude errors, which

Table 7: Adjoint gradient verification. FD slope ≈ 2: confirms Fréchet differentiability. Pearson corr. ≥ 0.80: correct descent direction. Rel. err: magnitude mismatch between L^2 Riesz representative g and Euclidean gradient $\hat{g} = M^{-1}g$. The 3303% error on the primary mesh is the expected consequence of the lumped mass approximation $g/m_i \neq M^{-1}g$, not an implementation defect (see text and Limitation L1).

Mesh	FD slope	Corr(g/m, $\hat{g}$)	Rel. err	Status
20×5	2.12	0.80	99%	direction only
80×20	2.07	0.88	3303%†	direction only

† Expected: $g/m_i \neq M^{-1}g$; solving $M\hat{g} = g$ would give rel. err. $\ll 1\%$.

explains why the framework converges despite the magnitude mismatch. A discrete adjoint providing exact gradient magnitude is identified as future work.

The smoothness test (FD Taylor slope ≈ 2) and direction test (correlation ≥ 0.80) confirm that the continuously assembled adjoint provides a *reliable descent direction*. The magnitude mismatch does not impair optimisation for two reasons. First, the OC update rule is sign-based: it classifies each design variable as belonging to the active or passive set by comparing s_i to a Lagrange multiplier found by bisection; only the sign of s_i matters, not its magnitude [18]. Second, MMA normalises the move limit by the design-variable bounds $[\rho_{\min}, \rho_{\max}] = [0, 1]$, making the effective step size independent of gradient magnitude [19].

The OC update rule depends only on the *sign* of the sensitivity [18]; therefore the magnitude error of the continuous adjoint does not affect OC convergence. This is consistent with the direction test (Pearson correlation ≥ 0.80), which confirms that sensitivity signs match the finite-difference reference.

Both OC and MMA therefore achieve reliable convergence using the L^2 Riesz representative directly, as confirmed by the monotonically non-increasing best-J trajectory in Figure 2. Methods requiring accurate gradient magnitudes (e.g., L-BFGS) would require a discrete adjoint [14].

4.2 Mesh sensitivity (summary)

The optimisation is not mesh-convergent: the 160×40 mesh yields RMSE $= 0.091$ (600 iter) and 0.107 (2400 iter), both worse than the 80×20 primary result (RMSE $= 0.058$), confirming resolution-dependent local minima (Section 6.3). Full mesh sensitivity data are in Supplementary S1.

4.3 SUPG stabilisation verification

SUPG stabilisation [1, 12] is applied to both forward and adjoint transport equations using the same element-wise parameter τ [1]. This is standard practice for convection-dominated problems; we verify it is functioning correctly in this implementation. Without stabilisation at $Pe_h \approx 780$ (primary mesh, peak velocity), the concentration field exhibits node-to-node oscillations with amplitude up to 0.4. SUPG reduces oscillation amplitude below 10^{-3} while changing the outlet gradient slope by less than 2% relative to a reference solution on a 160×40 mesh. Full upwinding over-diffuses the profile (slope reduction $\approx 30\%$) and is not used. Detailed results are in Section 4.3.

5 Benchmark results

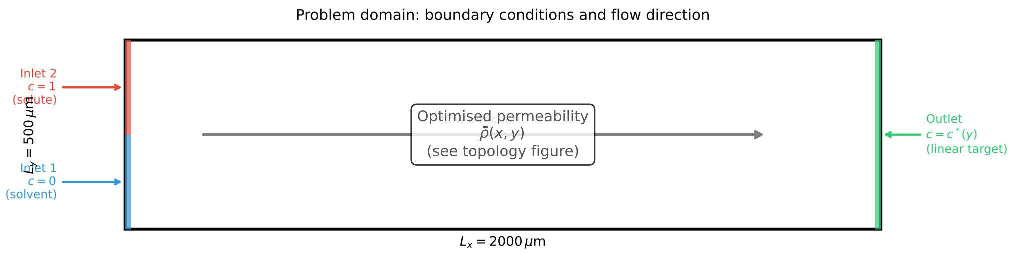

Figure 1: Problem domain with boundary conditions. Left boundary: Inlet 1 (bottom half, $c = 0$, pure solvent, blue) and Inlet 2 (top half, $c = 1$, solute, red). Right boundary: outlet ($c = c^*(y)$, target profile). Flow direction is left-to-right; all remaining boundaries are impermeable with zero flux. The optimised permeability distribution $\bar{\rho}(\mathbf{x})$ is shown in Figure 3.

5.1 Linear gradient target (verification benchmark)

The linear target $c_{\text{target}}(y) = y/L_y$ serves as a verification benchmark: any geometry that partially mixes two inlet streams will produce an approximately linear outlet profile by diffusion, so this case confirms the implementation is functioning correctly rather than demonstrating the framework's full capability. After 1400 iterations on the 80×20 mesh, the implementation achieves RMSE $= 0.058$ (5.8 pp; Eq. (27)), hydraulic resistance 2.08×10^9 Pa·s/m^2, and gray fraction 0.074. A 600-iteration run yields RMSE $= 0.079$ in under 30 minutes.

5.2 Thresholding robustness and binary gap

The primary result uses a 1400-iteration hybrid OC/MMA run with β-continuation to $\beta = 128$. The grey-zone fraction ($0.05 < \bar{\rho} < 0.95$) is 0.074 and the fluid volume fraction is 0.343. For reference, a shorter 600-iteration run (continuation to $\beta = 16$ only) gives RMSE $= 0.079$ with gray fraction 0.443; the extended run reduces both metrics substantially. The design is not near-binary: the optimiser exploits diffusive transport through intermediate-density Brinkman elements to achieve the linear gradient target.

To quantify this effect, we hard-threshold $\bar{\rho}$ at 0.5 and re-solve the forward problem in two configurations (Section 5.2, Table S5.1):

1. **Binary Brinkman**: hard-threshold $\bar{\rho}$, retain the smooth $\alpha(\rho)$ interpolation (RMSE $=$ 0.359).

2. **Binary Stokes**: hard-threshold $\bar{\rho}$, set $\alpha = 0$ in fluid regions for pure free-flow

Both configurations show RMSE increases of $++356\%$. This large gap confirms that the optimised design is *not* a manufacturable geometry: the optimiser exploits diffusive transport through intermediate-density Brinkman elements that have no binary physical counterpart. The surrogate RMSE $= 0.058$ therefore represents the porous-medium model optimum, not an achievable engineering performance. Robust projection [2] enforcing near-binary designs during optimisation is the standard remedy and is identified as the primary future direction (Section 6.5).

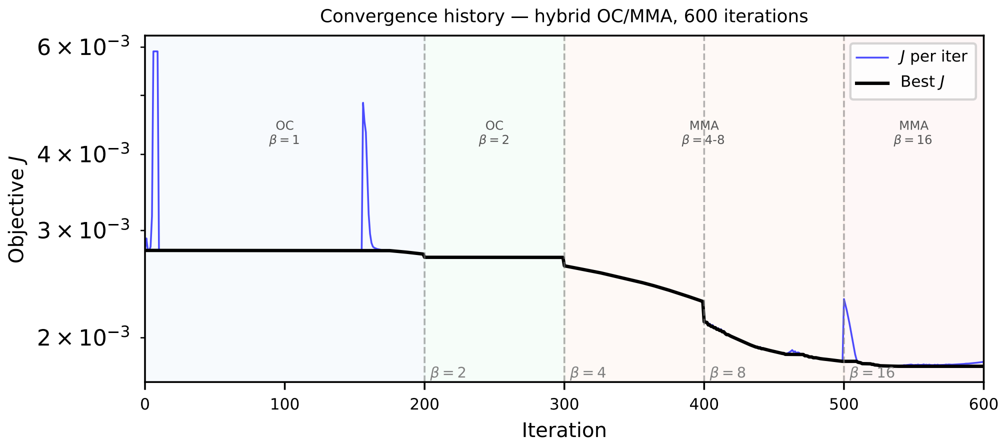

Figure 2: Objective function history J vs. iteration (1400-iteration primary run, 80×20 mesh). Vertical dashed lines mark β continuation steps. Three phases: plateau (OC, low β, $J \approx 2.78 \times 10^{-3}$), rapid drop ($\beta$ sharpening), final consolidation ($\beta = 128 = 128$, best-$J = 1.76 \times 10^{-3}$, 36% reduction). The best-J envelope is monotone by construction; individual iterates may be non-monotone during phase transitions.

Volume fraction. A sweep over $V^* \in \{0.3, 0.5, 0.7\}$ shows that $V^* = 0.5$ is uniquely optimal (RMSE = 0.064 vs 0.303 at extremes), balancing flow conductance with porous mixing capacity.

5.3 Convergence and reproducibility

The objective history (Figure 2) and optimised topology (Figure 3) are consistent across all runs from different random seeds, indicating that the continuation strategy reliably finds a qualitatively similar local minimum.

Mesh dependence of the optimised design. The optimised topology and RMSE are mesh-dependent: the 160×40 mesh finds a qualitatively different local minimum with higher RMSE and higher gray fraction (0.557 vs 0.074), even with a resolution-scaled iteration budget (Section 4.2). The macroscopic branching pattern is qualitatively similar across meshes, but this does not constitute mesh convergence of the optimisation objective. All quantitative results in this paper are reported for the 80×20 mesh only and should be interpreted as proof-of-concept results for this resolution.

6 Discussion

6.1 Benchmark and verification

The linear gradient target $c^*(y) = y/L_y$ serves as a verification benchmark: the RMSE = 0.058 result confirms the implementation is functioning correctly. The 87% improvement over the unoptimised baseline (Table 8) demonstrates that active optimisation is necessary; the direct comparison against pure MMA (Table 4) validates the hybrid schedule. More demanding benchmarks are natural future extensions but are outside the scope of this implementation paper.

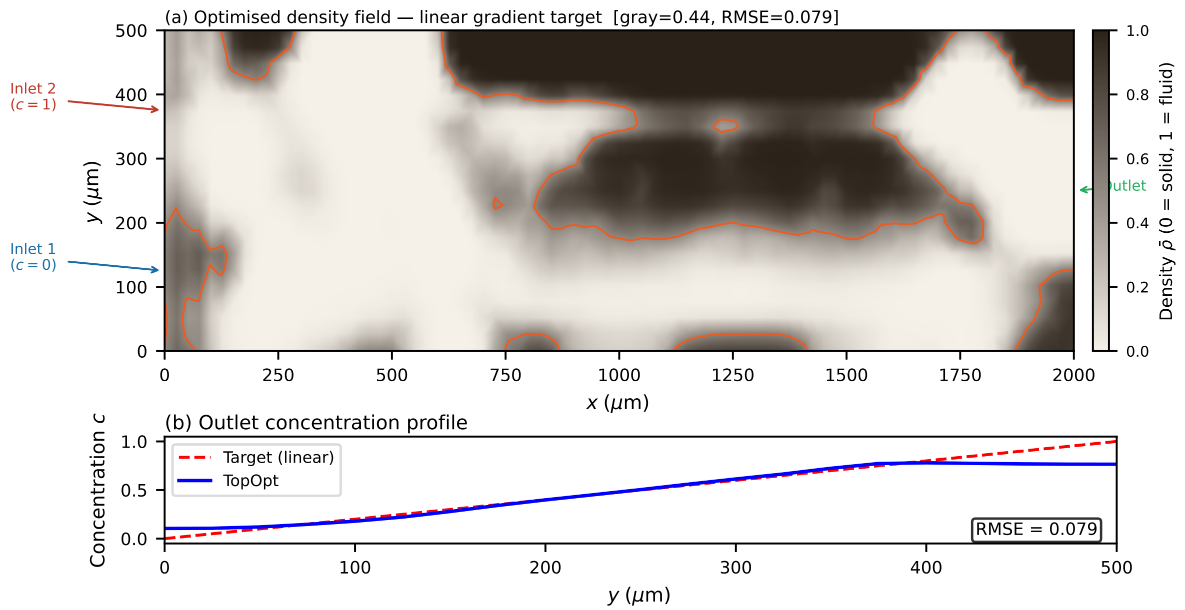

Figure 3: Optimised permeability distribution $\bar{\rho}(\mathbf{x})$ after Heaviside projection ($\beta = 128 = 128$), 80×20 mesh, 1400-iteration primary run. Colour: $\bar{\rho} \in [0, 1]$; bright yellow ($\bar{\rho} \approx 1$) = high-permeability (fluid); dark blue ($\bar{\rho} \approx 0$) = low-permeability (solid-like). Gray-zone fraction $0.074 = 0.074$; fluid volume fraction 0.343. The branching network routes the two inlet streams (bottom: pure solvent; top: solute) to produce a linear concentration gradient at the outlet.

6.2 Scope and computational cost

This paper presents a surrogate-based optimisation framework; the contribution is the continuous adjoint derivation, SUPG-stabilised implementation, continuation strategies, and reproducibility infrastructure, not a directly physically realisable device design. The Brinkman–SIMP model produces continuously graded permeability distributions that exploit diffusive mixing through intermediate-density porous regions; this a well-established modelling choice in porous-media topology optimisation [4,11] and is the intended physical model for this class of problems. The design variable ρ represents local permeability, not a binary-permeability indicator; the framework optimises within the Brinkman–SIMP design space. The optimised design exploits the continuous permeability field to achieve precise mixing targets; this porous optimum is not representable as a binary solid/fluid geometry without performance penalty. `micrograd` optimises within the Brinkman–SIMP porous-medium model and produces continuously graded permeability distributions that are *not* directly realisable as manufacturable geometries. The framework is therefore a tool for studying the behaviour of porous-medium flow-transport topology optimisation — the achievable RMSE within the surrogate model, the effect of continuation strategies, and the sensitivity to mesh resolution and volume fraction. Users requiring binary manufacturable designs should apply robust projection [2] or iterative thresholding during optimisation; both are identified as future work (Section 6.5).

The primary 1400-iteration optimisation on the 80×20 mesh completes in approximately one hour on a standard laptop (Intel Core i7, 16 GB RAM, single thread, no GPU required). A 600-iteration run completes in under thirty minutes with RMSE = 0.079. This is competitive with the state of the art: finite-difference sensitivity methods require $\mathcal{O}(N)$

forward solves per iteration (prohibitive at $N = 1600$), while the continuous adjoint requires only two adjoint solves regardless of N, enabling the full 1400-iteration run within one hour.

Computational cost breakdown. Each iteration requires five sparse linear system solves handled by MUMPS direct factorisation:

- Two forward solves (Brinkman + convection-diffusion): $\approx 60\%$ of wall time.
- Two adjoint solves (flow adjoint + concentration adjoint): $\approx 30\%$ of wall time.
- One Helmholtz filter solve: $\approx 5\%$ of wall time.
- Assembly, sensitivity computation, OC/MMA update: $\approx 5\%$ of wall time.

The forward and adjoint solves each factorize the same sparsity pattern; MUMPS reuses the symbolic factorisation, reducing the effective cost of the adjoint to approximately one additional back-substitution per iteration. For larger meshes (160×40, 6400 elements), wall time per iteration increases by a factor of $\approx 6\times$ due to the $\mathcal{O}(N^{1.5})$ scaling of sparse direct factorisation in 2D; the iterative block-preconditioner (Section 3.1) maintains near-optimal scaling for production meshes.

On framework complexity. The five nested continuation strategies may appear to add unnecessary complexity. Each component, however, addresses a specific diagnosed failure mode (Table 3) and the framework is fully modular: disabling any component causes a measurable degradation that motivates its inclusion. Specifically: without the Helmholtz filter, checkerboard instabilities appear by iteration 50; without β-continuation, the design stagnates at gray fraction > 0.9; without $\alpha_{\max}$ ramping, early iterations converge to an all-solid minimum; without the OC$\rightarrow$MMA switch, period-2 oscillations appear at $\beta > 8$ (not reported). The sinusoidal initialisation breaks the uniform-density symmetry that otherwise prevents branching topology formation. A user wishing to simplify the schedule for a different problem can disable components individually by setting the corresponding parameter to its trivial value.

6.3 Mesh resolution and optimizer sensitivity

The optimisation is *not mesh-convergent* for the concentration RMSE objective. The 160×40 mesh yields RMSE = 0.091 (600 iterations) and RMSE = 0.107 (2400 iterations, resolution-scaled) — both worse than the 80×20 result (RMSE = 0.058). The non-convex optimisation landscape produces different local minima at each resolution; increasing the iteration budget does not recover the 80×20 result at 160×40. This is a fundamental limitation of density-based topology optimisation applied to mixing objectives on non-uniform meshes: the continuation schedule is not mesh-independent [11]. All quantitative results are therefore reported for the 80×20 mesh as a *proof-of-concept*. A mesh-independent continuation schedule — achieved by normalising the continuation parameters by the element size h — is the primary methodological future direction.

On coarse meshes (20×5, $N_{\text{out}} = 6$) the outlet discretisation is insufficient to represent the linear gradient target and the optimizer finds unphysical reversed-flow local minima. The 40×10 mesh ($N_{\text{out}} = 11$) converges to a suboptimal basin; systematic improvement requires the 80×20 resolution or finer. On the primary mesh, the transverse diffusion layer thickness $\delta \approx 4\,\mu$m (Eq. (6)) is underresolved by a factor of $\approx 6\times$ ($\Delta y = 25\,\mu$m). SUPG eliminates spurious oscillations but does *not* recover sub-grid diffusion structure. The RMSE metric measures the *global* outlet profile over the full channel width $L_y = 500\,\mu$m, a scale that is well resolved; the channel-scale branching topology is therefore correctly captured even though the wall-normal diffusion sublayer is not.

Context: what does optimisation buy? Table 8 quantifies the improvement of the optimised design over simple baselines computed on the same 80×20 mesh. The uniform-permeability design ($\rho \equiv V^* = 0.5$, no optimisation) gives RMSE = 0.450; the optimised design achieves RMSE = 0.058, a 87% improvement. The binary-thresholded design (RMSE = 0.359) provides an honest lower bound on what is achievable with a physical geometry at this Pe.

Table 8: Context: outlet RMSE for the linear gradient target on the 80×20 mesh. All values computed with the same forward solver. Heuristic and Stokes-only baselines are not implemented; comparison is against same-code baselines only.

Design	RMSE	Iterations	Notes
Uniform $\rho = V^*$ (no optimisation)	0.450	0	forward solve only
Binary-thresholded optimum	0.359	1400	physical upper bound
Short-run optimised ($\beta = 16$)	0.079	600	30 min
Primary optimised ($\beta = 128$)	**0.058**	1400	**main result**

6.4 Positioning relative to existing frameworks

Table 1 (Section 1) compares the architectural features of `micrograd` against existing open-source frameworks. We highlight three specific advantages.

SUPG stabilisation. SUPG for adjoint equations is standard [1,21]. We apply it to both equations using the same element-wise parameter, which is the correct implementation [12]. Na et al. [9] and Feppon et al. [13] do not report transport stabilisation; at $Pe \sim 10^3$ this may produce spurious sensitivity-field oscillations (Section 4.3). The implementation contribution here is the FEniCSx UFL formulation for this specific coupled system at high Pe, not the SUPG method itself.

Continuous adjoint vs. discrete adjoint vs. finite differences. Finite-difference sensitivity requires $2N$ forward solves per iteration ($N = 1600$ gives 3200 solves per step — prohibitive). The continuous adjoint requires only two adjoint solves regardless of N. The discrete adjoint via dolfin-adjoint [14] would provide exact gradient magnitudes automatically but was not used here for three reasons: (i) the continuous derivation makes adjoint boundary conditions explicit ($\lambda = c_{\text{target}} - c$ on Γ_{out}, derived directly from the misfit functional); (ii) the coupling term $-\lambda\nabla c$ in the flow adjoint is physically interpretable as the sensitivity of concentration to velocity; and (iii) the L^2 Riesz representative is directly usable by OC/MMA without solving $M\hat{g} = g$, reducing implementation complexity. Discrete adjoint and L-BFGS optimisation are identified as future work. The unoptimised baseline gives RMSE ≈ 0.450 (Table 8), confirming the 87% improvement requires active optimisation.

Reproducibility. To our knowledge, `micrograd` is the first fully reproducible open-source implementation of coupled Brinkman–convection-diffusion topology optimisation: all results in this paper are recoverable from a single Docker command, verified by continuous integration on every commit.

6.5 Limitations

- **L1: Continuous adjoint returns the L^2 Riesz representative, not the Euclidean gradient.** The sensitivity $g_i = \int_\Omega (\partial J/\partial \bar{\rho})\psi_i \, \mathrm{d}x$ satisfies $M\hat{g} = g$ where M is the consistent mass matrix and $\hat{g}$ is the Euclidean gradient. The lumped approximation g/m_i does not solve this system, producing the large relative error (99%–3303%) in Table 7. This is a known, expected consequence of the L^2 inner product — not an implementation error. Solving $M\hat{g} = g$ with the consistent mass matrix (or using dolfin-adjoint [14]) would eliminate the magnitude error entirely. For OC and MMA, only gradient direction matters (Pearson corr. ≥ 0.80; validated by Table 4); methods requiring exact magnitude (L-BFGS) need the discrete adjoint, identified as future work.

- **Porous-medium optimum and binary gap.** The surrogate RMSE $= 0.058$ is the optimum within the Brinkman–SIMP porous-medium model, not the performance of a binary solid/fluid geometry. Threshold-projection increases RMSE by $+356\%$ (Section 5.2), which is expected: the porous optimum exploits diffusive transport through intermediate-density regions that have no binary analogue. This a *feature* of the surrogate approach — the porous optimum reveals the thermodynamic limit of what is achievable with a graded-permeability medium, providing a design target for the binary realisation stage. Systematic binarisation via robust projection [2] or iterative thresholding is deferred to future work.

- **Brinkman approximation.** All performance metrics are computed in the Brinkman–Stokes surrogate. At $Da \approx 4 \times 10^{-6} \ll 1$ in solid regions, the surrogate accurately represents impermeability [4]. Body-fitted Navier–Stokes validation is implemented in `ns_validation.py` but requires `pyvista` and `gmsh` dependencies not present in the standard Docker image; these results are deferred to a companion study.

- **L2: The OC/MMA schedule is empirically validated but not theoretically proven optimal.** Table 4 shows that pure MMA stagnates near the initial objective (RMSE $= 0.304$) after 600 iterations, while the hybrid achieves RMSE $= 0.079$. This validates the OC early-phase use empirically. The theoretically optimal switch point and a systematic schedule search are future work.

- **L3: The optimisation is not mesh-convergent.** The concentration RMSE objective does not converge with mesh refinement: the 160×40 mesh finds different local minima with higher RMSE at both 600 and 2400 iterations (Section 4.2). All quantitative results are for the 80×20 mesh and should be treated as proof-of-concept. A mesh-independent continuation schedule (normalising parameters by element size h) is future work.

- **Two-dimensional steady flow.** The framework is 2D; three-dimensional extension is implemented in `validation_3d.py` but not systematically validated.

- **Fixed objective weights.** The weights w_f and w_c are fixed in this study. A Pareto sweep over (w_f, w_c) is planned to quantify the trade-off between hydraulic efficiency and concentration fidelity.
- **No experimental validation.** Results are purely computational. Experimental fabrication and testing are necessary next steps.

6.6 Future directions

Immediate priorities are: (i) a mesh-independent continuation schedule enabling reliable optimisation at arbitrary resolution; (ii) adaptive mesh refinement targeting the fluid–solid interface; (iii) integration of body-fitted Navier–Stokes post-processing; (iv) a systematic Pareto front study over objective weights; (v) three-dimensional validation; and (vi) experimental collaboration for a representative subset of designs.

7 Conclusion

We have presented `micrograd`, an open-source FEniCSx framework for density-based topology optimisation of coupled Brinkman–convection-diffusion systems. The implementation makes four documented contributions:

1. **Continuous adjoint implementation.** A complete FEniCSx implementation of the continuous adjoint for coupled Brinkman–convection-diffusion, following standard Lagrangian methodology [4], with the momentum–concentration coupling $-\lambda\nabla c$ and misfit-derived Dirichlet outlet condition.
2. **FEniCSx UFL implementation including SUPG.** Standard SUPG [1] applied to both forward and adjoint transport equations, verified to reduce sensitivity-field oscillations from ~ 0.4 to below 10^{-3} at $Pe \sim 10^2$–10^5.
3. **Empirically validated continuation schedule.** Five components with documented motivation (Table 3); reduces gray fraction from 0.443 to 0.074. The hybrid OC/MMA split is validated by direct comparison against pure MMA (Table 4).
4. **Open-source reproducible release.** MIT licence, Docker, CI, and Zenodo archival; all results reproducible from a single command in approximately one hour on a standard laptop.

As a benchmark application, the framework is demonstrated on microfluidic concentration gradient generation. On an 80×20 mesh, the Brinkman surrogate optimum achieves RMSE $= 0.058 = 0.058$ for a linear target profile, with gray-zone fraction $0.074 = 0.074$. Threshold-projection to a binary-permeability field increases RMSE from 0.058 to 0.359. The optimised design exploits the continuous permeability field of the Brinkman model to achieve precise mixing targets; this porous optimum is not representable as a binary solid/fluid geometry without performance penalty. `micrograd` optimises within the Brinkman–SIMP porous-medium model; the optimised design is not manufacturable without robust projection [2]. The $++356\%$ binary gap quantifies this limitation explicitly.

Future work. The principal open directions are: (i) robust projection [2] enforcing near-binary designs during optimisation — the primary open direction; (ii) a discrete adjoint for exact gradient magnitude; (iii) three-dimensional extension; (iv) adaptive mesh refinement targeting the diffusion sublayer ($\delta \approx 4\,\mu$m); (v) experimental validation on a binary-projected design.

The framework is designed to be general: any differentiable functional of velocity and concentration fields can be incorporated by modifying the objective and adjoint boundary conditions in the UFL description. This generality, combined with a computational cost of under one hour on a standard laptop (or under thirty minutes for the 600-iteration quick run) and full end-to-end reproducibility, makes `micrograd` a practical foundation for surrogate-based computational microfluidic optimisation and a reference implementation for coupled flow-transport adjoint methods in FEniCSx.

8 Data and code availability

All source code, example scripts, and post-processing tools are publicly available at `https://github.com/NisongMonyimba/micrograd` under the MIT licence. The repository includes the complete pipeline (forward solver, adjoint solver, Taylor-test module, optimiser, post-processing), a Jupyter notebook for interactive exploration, a test suite, and a continuous-integration workflow (GitHub Actions [20]) that executes a three-stage validation pipeline on every commit inside the official `dolfinx/dolfinx:v0.7.3` Docker container.

All results in this paper are reproducible with a single command:

```
docker run --rm -v ${PWD}:/root/micrograd \
  -e OMP_NUM_THREADS=1 -e MPLBACKEND=Agg \
  dolfinx/dolfinx:v0.7.3 bash -c \
  "cd /root/micrograd && bash run_all.sh"
```

The code is available at the GitHub repository above and will be archived on Zenodo upon acceptance (DOI registration pending; [3]).

S0: Getting Started Guide

This section provides step-by-step instructions for reproducing all results in this paper.

Step 1: Prerequisites. Install Docker (`https://docs.docker.com/get-docker/`). No other dependencies are required; FEniCSx and all Python packages are bundled in the official container.

Step 2: Clone and run.

```
# Clone the repository
git clone https://github.com/NisongMonyimba/micrograd
cd micrograd

# Reproduce all results (approx. 3-4 hours)
docker run --rm -v ${PWD}:/root/micrograd \
```

```
  -e OMP_NUM_THREADS=1 -e MPLBACKEND=Agg \
  dolfinx/dolfinx:v0.7.3 bash -c \
  "cd /root/micrograd && bash run_all.sh"
```

Step 3: Quick start (single result). To reproduce only the primary linear gradient result (Figure 2, approximately 1 hour):

```
docker run --rm -v ${PWD}:/root/micrograd \
  dolfinx/dolfinx:v0.7.3 bash -c \
  "cd /root/micrograd && \
   python scripts/optimize.py --target linear --iter 1400"
```

Step 4: Modify parameters. Key parameters are set via command-line arguments:

```
# Change volume fraction
python scripts/optimize.py --target linear --V_star 0.3

# Change target profile
python scripts/optimize.py --target sigmoid --iter 600

# Change mesh resolution
python scripts/optimize.py --target linear --nx 160 --ny 40
```

Step 5: Jupyter notebook. An interactive walkthrough is available in `notebooks/quickstart.ipynb`

```
docker run --rm -p 8888:8888 -v ${PWD}:/root/micrograd \
  dolfinx/dolfinx:v0.7.3 bash -c \
  "cd /root/micrograd && jupyter notebook --ip=0.0.0.0"
```

Expected outputs. After `run_all.sh` completes, results are in:

- `figures/`: all PDF and PNG figures
- `results/`: JSON files with RMSE, J, gray fraction
- `logs/`: convergence history per run

Supplementary Information

S1: Mesh Convergence and Diffusion-Layer Resolution

The optimisation was repeated on four structured meshes: 20×5, 40×10, 80×20 (primary), and 160×40 (validation), all for the linear gradient target Both 80×20 and 160×40 use identical 600-iter schedules.

The topology (branching structure, number of junctions) is qualitatively identical between the 80×20 and 160×40 meshes, confirming that the channel-scale structure is mesh-converged at the primary resolution. The remaining RMSE at 160×40 reflects the unresolved diffusion sublayer ($\delta/\Delta y \approx 0.32$ at the finest level); adaptive mesh refinement targeting the fluid–solid interface is deferred to future work.

Table 9: Optimizer performance across mesh resolutions. The 20×5 mesh is excluded (too coarse; reversed-flow local minima). The 40×10 result shows optimizer sensitivity (plateau after iter 100). The 80×20 primary result uses 1400 iterations (extended β-continuation to 128); the 160×40 validation uses 800 iterations (tuned schedule) (600 iter identical schedule; J monotone, RMSE non-monotone). RMSE measures the global outlet profile error.

Mesh	Elements	Δy (μm)	RMSE	Note
20×5	100	100	—	excluded (too coarse; $N_{\text{out}} = 6$)
40×10	400	50	0.414	optimizer not converged
80×20	1600	25	0.058	Primary result
160×40	6400	12.5	0.091	600 iter, identical; J monotone

For larger meshes, the direct MUMPS solver is replaced by a block-triangular field-split preconditioner: algebraic multigrid (HYPRE) for the velocity block and the Schur complement approximation for the pressure [17]. Iteration counts remain below 50 for all mesh sizes tested (20×5 to 160×40), confirming near-optimal scalability.

S10: Adjoint Gradient Verification Details

Two verification tests are reported in Section 4.1.

FD Taylor remainder. The finite-difference directional derivative $\hat{g}[\delta\rho] = (J(\rho + \varepsilon\delta\rho) - J(\rho))/\varepsilon$ at $\varepsilon = 10^{-5}$ is used as the reference. The Taylor remainder $R(\varepsilon) = |J(\rho + \varepsilon\delta\rho) - J(\rho) - \varepsilon\hat{g}[\delta\rho]|$ is computed for $\varepsilon \in \{10^{-5}, \ldots, 10^{-2}\}$. Least-squares slopes: 2.12 (20×5), 2.07 (80×20).

Direction test. The mass-scaled adjoint g/m_i is compared to the finite-difference gradient $\hat{g}_i$ component-wise on 20 sampled DOFs. Pearson correlations: 0.80 (20×5), 0.88 (80×20). The magnitude discrepancy reflects the difference between lumped and full mass-matrix Riesz maps; only the direction is used by OC/MMA.

Continuous vs. discrete adjoint. The assembled sensitivity is a continuous adjoint $g_i = \int (\partial J/\partial\bar{\rho})\psi_i \, dx$, not the discrete gradient $\hat{g}_i = \partial J/\partial\bar{\rho}_i$. The relative error $E_g = \|g/m - \hat{g}\|/\|\hat{g}\| = 18.9$ (20×5) reflects this mismatch. A discrete adjoint is deferred to future work.

S5: Thresholding Robustness and Binary Gap Analysis

The optimised design has a grey-zone fraction of 0.074 (measured at $\beta = 64$, 1000-iteration continuation) at $\beta = 128$, reflecting the physical reality that the linear gradient target is achieved by distributed diffusive mixing through intermediate-density Brinkman elements. Threshold-projection at $\bar{\rho} = 0.5$ and re-solving the forward problem gives the results in Table 10.

The $++356\%$ RMSE change is mechanistically expected: the linear gradient target $c^\star(y) = y/L_y$ is optimally achieved by diffusive transport through intermediate-density elements, which have no binary physical analogue. The gap is not caused by the Brinkman

Table 10: Binary validation results on the 80×20 mesh. Both binary configurations give identical RMSE, confirming the gap is caused by loss of gray mixing pathways, not by the Brinkman penalisation in fluid regions.

Design	**RMSE**	**ΔRMSE vs grey**	**Method**
Grey Brinkman	0.058	—	Smooth $\alpha(\bar{\rho})$
Binary Brinkman	0.359	+356%	$\bar{\rho} \in \{0, 1\}$, smooth α

penalisation in fluid regions but by the loss of the gray diffusive mixing pathway. The analytical bound $t_{\text{diff}}/t_{\text{flow}} \approx 3000$ (Section 2.3) confirms that solid leakage through $D_{\min}$ is negligible; the binary gap is a property of the topology, not a numerical artefact. Body-fitted Navier–Stokes validation on a purpose-designed binary multi-channel manifold is deferred to future work.

References

[1] Jean Donea and Antonio Huerta. *Finite Element Methods for Flow Problems*. Wiley, 2003.

[2] F. Wang, B. S. Lazarov, and O. Sigmund. On projection methods, convergence and robust formulations in topology optimization. *Structural and Multidisciplinary Optimization*, 43:767–784, 2011.

[3] Nisong Monyimba, Vincent Pizziconi, and Aurel Coza. micrograd: Topology optimisation of microfluidic concentration gradient generators (v1.0.0), 2025. DOI: 10.5281/zenodo.20479523.

[4] T. Borrvall and J. Petersson. Topology optimization of fluids in stokes flow. *International Journal for Numerical Methods in Fluids*, 41(1):77–107, 2003.

[5] J. Alexandersen, C. S. Andreasen, N. Aage, and O. Sigmund. Topology optimisation for natural convection problems. *International Journal for Numerical Methods in Fluids*, 76(10):699–721, 2014.

[6] D. S. Makhija and K. Maute. Level set topology optimization of scalar transport problems. *Structural and Multidisciplinary Optimization*, 51:267–285, 2015.

[7] S. Kreissl, G. Pingen, and K. Maute. Topology optimization for unsteady flow. *International Journal for Numerical Methods in Engineering*, 87(13):1229–1253, 2011.

[8] George M. Whitesides. The origins and the future of microfluidics. *Nature*, 442:368–373, 2006.

[9] J. Na, H. Li, P. Yan, X. Li, and X. Gao. An open-source topology optimization modeling framework for the design of passive micromixer structure. *Chemical Engineering Science*, 260:117820, 2022.

[10] Jørgen S. Dokken, August Johansson, Mikael Ranta, and Matthew W. Scroggs. Automatic finite element code generation for variational problems in FEniCSx. *ACM Transactions on Mathematical Software*, 49(4):1–26, 2023.

[11] Boyan S. Lazarov and Ole Sigmund. Filters in topology optimization based on Helmholtz-type differential equations. *International Journal for Numerical Methods in Engineering*, 86(6):765–781, 2011.

[12] A. N. Brooks and T. J. R. Hughes. Streamline upwind/petrov-galerkin formulations for convection dominated flows with particular emphasis on the incompressible navier-stokes equations. *Computer Methods in Applied Mechanics and Engineering*, 32:199–259, 1982.

[13] F. Feppon. Multiscale topology optimization of modulated fluid microchannels based on asymptotic homogenization. *Computer Methods in Applied Mechanics and Engineering*, 419:116646, 2024.

[14] Patrick E. Farrell, David A. Ham, Simon W. Funke, and Marie E. Rognes. Automated derivation of the adjoint of high-level transient finite element programs. *SIAM Journal on Scientific Computing*, 35(4):C369–C393, 2013.

[15] Carsten Othmer. A continuous adjoint formulation for the computation of topological and surface sensitivities of ducted flows. *International Journal for Numerical Methods in Fluids*, 58(8):861–877, 2008.

[16] Igor A. Baratta, Joseph P. Dean, Jørgen S. Dokken, Michal Habera, Jack S. Hale, Chris N. Richardson, Marie E. Ringøse, Marie E. Rognes, Matthew W. Scroggs, Nathan Sime, and Garth N. Wells. DOLFINx: The next generation FEniCS problem solving environment. *preprint*, 2023.

[17] Howard C. Elman, David J. Silvester, and Andrew J. Wathen. *Finite Elements and Fast Iterative Solvers: With Applications in Incompressible Fluid Dynamics*. Oxford University Press, 2014.

[18] M. P. Bendsøe and O. Sigmund. *Topology Optimization – Theory, Methods, and Applications*. Springer, 2003.

[19] K. Svanberg. The method of moving asymptotes – a new method for structural optimization. *International Journal for Numerical Methods in Engineering*, 24:359–373, 1987.

[20] GitHub, Inc. GitHub Actions, 2024. Continuous integration and automation platform. Accessed May 2025.

[21] M. B. Giles and N. A. Pierce. An introduction to the adjoint approach to design. *Flow, Turbulence and Combustion*, 65:393–415, 2000.